



INFRANOR GROUP COMPANY

CYCAD

User Guide

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Important:

This notice explains normal and standard programming operations for the numerical control. In view of the fact that numerical controls can be equipped with configurable functions by the press manufacturer for his own specific purposes, please refer to the manufacturer-supplied complementary instructions regarding the programming of these functions.

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SAFETY AND MAINTENANCE INSTRUCTIONS



- The operator must be trained for working with the machine on which the numerical control is installed.
- Improper use of the numerical control can cause heavy damage on equipment and/or injuries to people.
- Modification of machine parameters can cause important material damage or lead to irregular product quality.
- The rear panel may only be removed by a qualified technician (danger of electrocution).
- Do not expose the numerical control to excessive humidity so as to avoid any risk of electrocution and any deterioration of the equipment.
- Make sure the numerical control is disconnected from the mains power before carrying out any cleaning. Do not use liquids based on alcohol or ammoniac.
- In case of malfunction of the numerical control, call a technician.
- Do not expose the numerical control to direct sun rays or any other heat source.
- Do not place the numerical control in the neighbourhood of magnetic equipment such as transformers, motors or devices which generate interference (welding machines, etc.)
- Replace fan filters at regular intervals so as to avoid overheating.

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ABOUT THIS MANUAL

This manual describes the CYCAD option which allows the PC1200 to read and convert DXF and IGES files for use on PC1200.

This manual addresses operators having a working knowledge on CAD systems. It is necessary for these persons to also be familiar with the PC1200 software, to know how to create products with this software and to have read the PC1200 user guide.

This manual can evolve, it is you, the operators, who can help us to help you more. If you have any remarks concerning this manual, please write to us at:

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INTRODUCTION

The CYCAD option offers conversion possibilities between DXF or IGES files generated by a **CAD** programme such as Autocad, CADKey, HiCad etc and files accepted by the numerical controls DNC 80/800/900/1200 from CYBELEC (in both directions).

The DXF (IGES) file describes the product in its entirety so that after conversion the DNC will be able to fully exploit that product. Use of the CYCAD conversion program, in conjunction with a **CAD** program, will bring about the following gains:

- You can use an existing drawing directly on a DNC, without having to re-programme the product.
- It is possible to include such attributes as holes and circles on the drawing - a feature not supported by the DNC.
- You can employ all the familiar CAD techniques to develop your drawing.
- Carrying out modifications on a drawing will be easier than on the DNC. For example, you can re-shape the product even after the bends have been defined.

Examples

Several example files in the DXF and DWG formats are available on the CYCAD diskette. Should this diskette be missing, do not hesitate to request it from our After Sales Service.

Allow yourself some time to open these files with your CAD system and with PC1200 / CYCAD. In each example, one or several particular cases are shown in detail. You will find the names of the example files in the sections where each particular case is mentioned.

WORKING METHOD

1. You design the product on your CAD system, according to the conventions explained below.
2. You save it to a DXF (IGES) format file.
3. You start the PC1200 software program. In the **INTERFACE CAD** page, you read, modify and convert the DXF (IGES) file to the "CYBELEC" format.
4. You may now modify the product by means of the PC1200 functions, add missing information if necessary, test the feasibility of the product and then transfer it to the numerical control.

You can also convert an existing file (.DAT file) into a **DXF (IGES)** file that can be used with a **CAD** program.

The drawing conventions for the **DXF (IGES)** file are strictly the same for both conversion directions.

The CYCAD option allows two different working approaches:

conversion of "untreated" CAD files and

conversion of "prepared" CAD files.

These two possibilities with their advantages and inconvenients are summarized hereafter.

CONVERSION OF "UNTREATED" CAD FILES

CYCAD is capable of reading "untreated" DXF files, i.e. files that have been converted in the state in which they have been created in the CAD system, without the conventions described hereunder being applied.

From the **CAD INTERFACE** page of the PC1200 software, the program filters certain unwanted information and displays the contents of the CAD file which will need to be "purified" and in which it will be necessary to select the elements needed by PC1200 (see *Use of CYCAD* section, page 25).

Existing CAD files

The advantage is that existing DXF or IGES files can be converted without having to do any work on them at CAD level. This is particularly convenient when the files have been generated on a system which is not available in the company.

However, such a file must be a 2D file, for CYCAD does not recognize 3D files. Furthermore, the file must contain at least the outline of the product and respect a certain number of limitations described in the *CAD Drawing Conventions* section, page 9.

CONVERSION OF "PREPARED" FILES

If the drawing conventions have been respected during creation of the products on the CAD system, the CYCAD converter is able to automatically insert the information at the correct locations (see *CAD Drawing Conventions*, page 9).

Conversion speed

Preparation of the CAD file may seem a bit long because the drawing conventions must be respected, but the conversion is greatly facilitated. The operator will need to spend much less time (if any) in the **CAD INTERFACE** page of PC1200, and the product can be "taken over" by PC1200 much faster.

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CAD DRAWING CONVENTIONS

The drawing is organized into various levels, of which just three are relevant to CYCAD.

- The first pre-defined level describes the overall shape of the product, with general data on it. This is called **OUTLINE**.
- The second pre-defined level defines the bends and associated data. This is called **BENDS**.
- The third (optional) pre-defined level contains section and order of bending data, together with support details. This is called **SECTIONS**.

You may use the other levels for your own needs. The conversion program simply ignores the levels which are not needed. It is recommended to assign particular colors or line types to the various levels. An example file treated this way can be found on the CYCAD diskette. The file names are EX_CY.DWG (for AutoCAD) and EX_CY.DXF.

At this stage it is helpful to remember that the notion of sectioning a product is not critical when designing it. It is not necessary to indicate section positions, since they are generated automatically by the conversion programme. If however the section location becomes important (eg. in determining the support position) you can in fact indicate it.

Hint

CYCAD is capable of exporting products created on PC1200 in the DXF or IGES format. It is thus easy for you to create yourself example DXF or IGES files respecting the CYCAD conventions and export a product with DXF or IGES format to look at them in your CAD system. See *Exporting a File in DXF or IGES Format*, page 30.

GENERALITIES

Texts:

The character type (font) and the size of the characters is unimportant.
When the position of the text is relevant, the lower left corner of the text is taken into account.

THE "OUTLINE" LEVEL

The text **CYCAD : OUTLINE** (in upper case characters) must be present for the level to be correctly identified.

PRODUCT OUTLINE

Arcs

The product shape must be defined as a series of lines and connected arcs. The outline of the product must be entirely closed (no open ends). The shape will correspond to that drawn on the **PRODUCT FLAT** page of PC1200.

In the moment of conversion, the arcs are converted into a series of segments. The number of segments corresponding to an arc can be specified by the user, as indicated in the *Setting Parameters for CYCAD / PC1200 Option* section, page 17.

The order in which the lines and arcs are drawn is unimportant; they must be simply end-to-end in the sense that each "summit" on the drawing must correspond to the junction of two lines/arcs. The total of these lines/arcs must not exceed 300 before conversion.

INTERNAL CUTS

You may draw cut-outs inside of the product outline in the same way as the outline, by means of lines and connected arcs. The external outline as well as the cuts must be closed shapes. It is, of course, necessary to take into account the max. number of segments (300).

An example of internal cuts can be found in the EXAMPLE1.DXF file. You can convert this file and see the result in PC1200.

If the the internal cut-outs are not really necessary for the comprehension or for the recognition of the drawing, and if they are not to be "worked on" (bend within the internal cut), it is recommended to put them into a non-CYCAD layer so that they will not appear in the product destined to PC1200.

PARAMETERS

Optionally, it is also possible on the **Outline** level to indicate various PC1200 parameters. You may specify just those values useful to you and omit the others. Each parameter is defined by an independent text, and the position of the text is unimportant. To define a parameter, write the corresponding keyword, the character = preceded and followed by a space and its value, which can either be a number or a text, according to the parameters.

- For real numbers (REALS) you may use the "minus" sign and the decimal point.
- "Whole" numbers (INTEGERS) are composed of figures only (0 - 9), including, if necessary, the minus sign.
- The length of the texts is limited to **25** characters.

The list of usable parameters and their associated keywords is given below:

Do not use any spaces inside the keywords. As an example, it is not acceptable to write **D I N** instead of **DIN**.

BHead = <text>	For folding presses only: corresponds to the name of the folding tool.
DefAngle = <real>	Defines a default angle for any angles that have not been specified in the Bends level.
Die = <text>	Name of the die, or of the lower tool in the case of a folding press.
DIN	Indicates that dimensions are 'post bending'.
Inch	Indicates that dimensions are expressed in inches.
Name = <text>	Product name.
Punch = <text>	Die name, or name of the upper tool in the case of a folding press.
Radius = <real>	Internal bending radius.
ScaleX = <real>	Horizontal scale factor (1.0 = no deformity).
ScaleY = <real>	Vertical scale factor.
Sigma = <real>	Sigma (...).
Thick = <real>	Thickness of the metal sheet.

Example:

Thick = 2.00
Punch = P12-3-30
Die = M10-30-1
etc.

Note: the text items must be placed on a vertical line, as shown in the above example.

Example

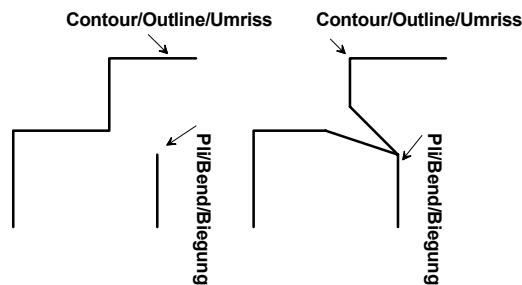
To obtain an example on how these parameters are used, export a file with DXF or IGES format and look at it with your CAD system. You can also see the EX_CY.DXF file on the CYCAD disquette.

THE "BENDS" LEVEL

The text **CYCAD : BENDS** (in upper case characters) must be present.

This level must include product bends in the form of a set of lines.

- Simple bends:** The bend line must intersect with the edge of the product in 2 places (not more, not less). You may use outline points as terminating points for bends. If a line has to cut a component of the outline, its endpoint can either be on the outline itself or can overshoot.
- Forced bends:** The bends can cut the outline at several points. All bends thus defined will effectively be actioned simultaneously. It is equally possible to define forced bends with unjoined lines. For this the lines must be aligned and 'barred' evenly (see example "EXEMPLE2.DXF"). Note: This is possible only for vertical or horizontal bends.
- Internal bends:** Bends which do not cut the outline but are near of the terminating points of it (at a distance smaller than 2 % of the largest dimension of the product), are treated differently. In order to be integrated into the product, the program modifies the outline by creating additional points to join the bending line (see example "EXEMPLE3.DXF"). In order to perfectly understand this example, it will be useful to convert EXEMPLE3.DXF and to view the result in the 3D construction page (**MODIFY 3D**). Also see the illustration below.



Drawing not at scale

- Max number of bends** The maximum number of bends allowed is **35, less the number of sections**. Furthermore there must not be more than 15 distinct orientations for these bends. You may thus have a maximum of 15 groups of parallel bends.

Angle values

Optionally, you may also indicate on this level the various bend angles. Certain bends can be left unassigned to particular angle values.

The angle values are indicated, as in the case of **OUTLINE**, with the aid of texts. These texts amount in fact to numbers (not keywords) and they must appear approximately in the middle of each bend, also in the case of a forced bend. Each value is recognized as belonging to the bend which is physically closest to the leftmost extremity of the text.

You are advised to avoid the use of over-large characters so as not to overfill your drawing. You may also indicate for each bend its associated tool (punch and die) and the internal radius. To this end you write after the angle:

C=<integer>	Ideal curve (number of bends)
D=<text>	Name of the die or, for a folding press, name of the lower tool
P=<text>	Name of the punch or, for a folding press, name of the upper tool.
R=<real>	Internal bending radius.
H=<text>	For a folding press only: corresponds to the name of the folding tool.
I	(I for India) For folding presses only: designates an inversed bend.

You may write partially or completely this set of fields and in any order. The field delimiting character is the '/'.

No space must be inserted after the "equal" sign.

You may also write the instructions in several separate texts for each bend.

Example: 90.5 / D=M1 / R=3.1 / C=5 / P=P1

This stream of data indicates that the bend's angle is 90.5 degrees, that it uses die M1, punch P1 has a radius of 3.1 mm and that there are 5 units of ideal curve.

THE "SECTIONS" LEVEL

The text "**CYCAD : SECTIONS**" (in upper case characters) must be present.

Sections: You can indicate at this level the relevant sections for a product.
Each section is defined by a line which intersects a component of the outline at two places. Bends perpendicular to this section, and crossing it, belong to this section, and no other.

Bend and Support Order

You can also specify at this level the bend order and support positions with the aid of texts. To do this you write close to the bend's mid point the letter '**b**', followed by a number (eg. **b1**, **b2** & **b3**). You should begin with number 1, and subsequently continue in exact sequence, leaving no gaps - for example going from 5 to 7.

The support is indicated by writing the letter '**s**' followed by the same number as the corresponding bend either near the middle of an existing bend or near the middle of a segment of the product outlinef (e.g. **s1**, **s2**, **s3**). For machines equipped with front gauges, the contact point with the front gauge must be indicated with the letter '**e**' followed by the number of the corresponding bend (e.g. **e1**, **e2**, **e3**).

You are not obliged to indicate the sequence number of every bend, but each time you do, you must also indicate its support (see example EXAMPLE4.DXF).

SUPPORTED ENTITIES

DXF:	ARC	
	LINE	
	LINE3D	
	POLYLINE	
	LWPOLYLINE	
	TEXT	
IGES:	CIRCULAR ARC	(100)
	LINE	(110)
	GENERAL NOTE	(212)

LIMITATIONS

- **1000** outline segments (including internal cuts, and conversion of curves into polylines)
- **30** sections
- The maximum number of bends is **130, less the number of sections**

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SETTING PARAMETERS FOR CYCAD / PC1200 OPTION

Before using the CYCAD option, you must verify that your protection key contains the options number 2, 5, 7 and 25. Should option 2 be missing, your key has not been programmed for the use of CYCAD. In this case, contact your supplier. It is also possible to verify the options of the protection key by calling up the **WELCOME** page of the PC1200 software. Only if the key and the option appear in this page you can assume that they have been recognized by the software. If the key is not recognized at all by the system, you cannot leave the Main Menu page, and the message **Illegal Configuration** is displayed.

ACTIVATING THE CYCAD OPTION

For the CYCAD program to be used, it is not sufficient to have a protection key with the CYCAD option. It is necessary to activate it in order to access the **CAD INTERFACE** page.

- For this, call the Machine Parameters page, then **OPTIONS**.

MACHINE PARAMETERS	OPTIONS
02 CAD Interface page - 03 Communications LINK7000	

- Select the sign - of **02 CAD Interface page**.
- Click on it so as to make appear the + sign.

The option is now validated.

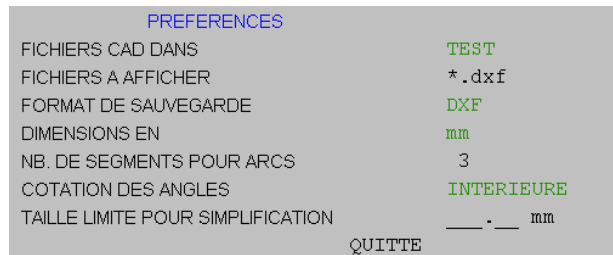
Should the choice **CAD INTERFACE** not appear in the Menu page, this means that the CYCAD option is not available in the protection key (see above).

CONFIGURATION OF PERIPHERALS

MACHINE PARAMETERS		PERIPHERALS		
PROGRAMMED PERIPHERALS				
N°	NAME	TYPE	PATH	OPT. NB
1	INTERNAL	DISK	P:\PC900\RAM\	200
2	FLOPPY	DISK	A:\	_____
ADDITIONAL PERIPHERALS				
▲	3	TOD01	DISK	P:\TOD01\
▼	4	DONE1	DISK	P:\DONE1\
	5	DXF	DISK	P:\DXF\
	6	IGES	DISK	P:\IGES\
	7	_____	DISK	_____

The paths in which can be found the DXF and / or IGES files must be defined within the machine parameters and in the **PERIPHERALS** page.

CONFIGURATION OF THE CAD INTERFACE PAGE



To reach the **PREFERENCES** page:

- Call up the **CAD INTERFACE** page.



- Click the icon, then icon.

CAD files in:

with right click, select in the appearing list the peripheral where the CAD files by default (TEST in this example) are located.

Files to display:

allows to insert a filter in the window for listed files. *.* display all files, regardless of the extension, *.DXF only lists DXF files etc.

Save format:

defines the format (DXF or IGES) in which the converted files (coming from PC1200) must be saved.

Dimensions in:

defines the measuring unit (mm or Inches).

Nb. of segments for arcs:

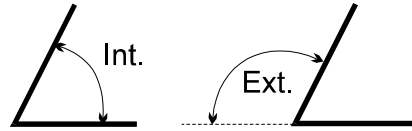
specifies into how many (straight) segments the arcs are to be transformed. If not programmed, this value will be 1. To see the effect of this feature, set this parameter to 3 or 5 and import the file called EX_CY.DXF. Each of the two large notches will be terminated by 3 or 5 segments respectively.

Be careful not to use too big values (max. 9) because for a complex part, the maximum number of segments accepted for the outline could be reached (see the *Limitations* section, page 15).

Angle dimensioning:

The choice is **internal** or **external**.

This indicates to the CYCAD converter in which way the angle values have been entered in the CAD drawing. This possibility only applies to CAD drawings. A mixed solution with internal / external angle values is not allowed.



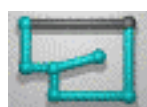
The PC1200 program only works with "internal" angle values.

**Simplification
limit size:**

This is the maximum distance of the selected segments for the simplification to be possible. See the Simplification icon, and the simplification example contained in the *Using CYCAD* section, page 25.

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DEFINITION OF CYCAD ICONS



Selection

Click on the element to select it.



Select polyline

Clicking on one of the segments selects all the lines of the "polyline" element.



Selection text

Click on the text (near lower left corner) to select it.



Select window

Selects all elements located inside a square created in the same way as the zooming function (note: the texts touched by the frame are also selected).



Select all

Selects (and de-selects) all elements of the drawing.



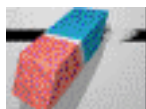
Transfer

Moves the selected elements to a different level.



Copy

Copies the selected elements to a different level. Be careful when using this feature, for if the copy has been carried out twice, there will be conversion problems. It is difficult to detect and to delete two elements which are superimposed on the same level. It is preferable to use the transfer function.



Eraser

Deletes the selected items.



Undo

Cancels the last action.



Zoom

Activate the zoom function, then click on the upper left corner of the desired portion, release the button, "drag" the cursor to the bottom right corner of the zone to be enlarged and click.

Click on the  icon to display the entire drawing.



Global view

Displays the entire drawing (after having used the zoom function).



Simplification

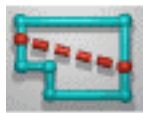
Replaces the small segments by one or several larger ones.

**Information**

Displays the CYCAD configuration window.

**Change icons**

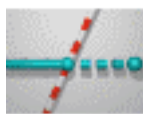
Displays the other **icons**.

**Create segment**

Creates a straight segment between 2 lines.
A click close to a point creates the end of the segment on this point. The segment is created in the same level as that of the first line.

**Lengthen segment**

Click on a line which becomes the support straight line.
Click on another line. This line is lengthened, so that it cuts the support straight line.

**Trim two crossing segments**

Click on a segment which becomes the support. Click on the segment which crosses. The selected part of the 2nd segment is eliminated up to the intersection with the support segment.

Selection

The selection of an object such as a line, polyline, text, etc. can be made by clicking on or close to the object. Clicking on the selected object de-selects it. It is possible to make multiple selections. A selected object appears in grey color.

The two selection icons are available in the two **CAD INTERFACE** pages, so as to avoid "back and forth" movements between the two pages (as from SJAFCIx software, mid-June 1997.)

LEVELS

The levels (or layers) recognized by CYCAD are represented by rectangles located on the top of the page.....



cyan

Outline

Contains the outline of the product, and general information on the product.



yellow

Bends

Contains the bending lines and indications for the bends.



red

Sections

Contains the sections, and indications for the bending order and supports.



black

Auxiliary functions

Useful for temporarily copying / transferring information.



blue

All others

Contents of all other levels.

The **Levels** field allows to activate either one of these levels (except for the level **All others**).

The active level is marked with a blue rectangle around the corresponding

rectangle , and its name appears in the Level field, as shown below.



In this example, the active level is **Outline**. The figure between brackets indicates the number of segments contained in the active level.

Masking a level

It is possible to mask one or several levels.

To this, just click inside the rectangle of the level to be masked. A masked level can be recognized by the black color inside the corresponding square



Caution: masked levels will not be converted.



In this example, the levels **Auxiliary** and **All others** are masked. The **Outline** level is active.

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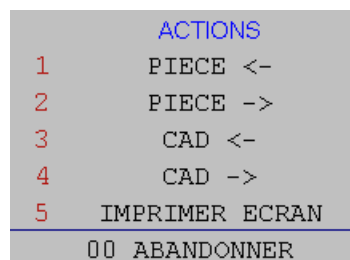
USING CYCAD

As seen in the *Working Method* section, it is possible to import "untreated" files or files "prepared" for CYCAD.

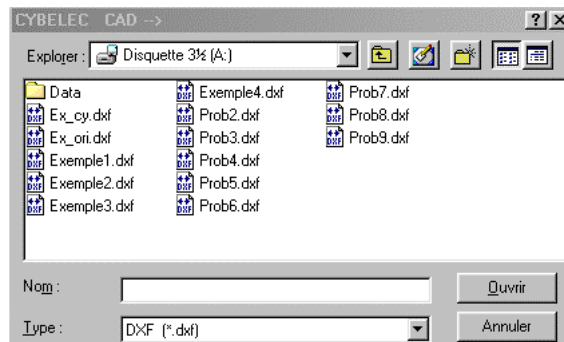
Hereafter, the import procedure for "untreated" files is explained because this procedure needs the most handling at CYCAD level. "Prepared" files can be imported exactly in the same fashion; they simply need less work to be done at CYCAD level.

General Procedure

- Click on the **ACTIONS** menu or press function key **F5**.



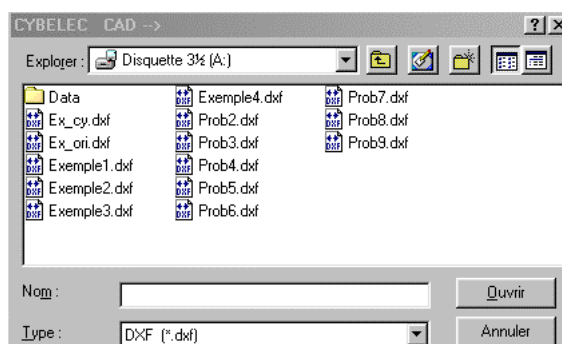
- Import the unprepared DXF file (**CAD →**).
- The following window appears, allowing you to navigate in the different directories of your system.



- Select the outline and move it to the **Outline** level.
- Select the bends and move them to the **Bends** level.
- If other indications, such as angle values, tools, material etc. are available, and provided that they are comply with the CYCAD prescriptions, you can also move them to the corresponding levels.
- If necessary, simplify the outline of the product in the **Outline** level.
- Convert the file into a PC1200 product (**PRODUCT ←**).
- In the PC1200 program, you can, if necessary, add missing information such as material, tools, tool position, etc.. Then you test the feasibility of the product as you would do with any other product created directly with PC1200.

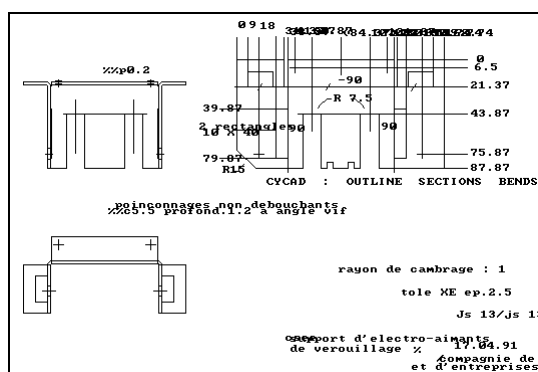
You will notice that this file does not have any specific level, nor any particular text corresponding to CYCAD standards. However, it contains the bending lines with the values of their angles and - this is the only CYCAD-specific item - three small oblique lines across the bending line of the upper part of the product, indicating that this will be a forced bend (see further on in this manual).

- Configure the peripherals and the **PREFERENCES** page of CYCAD as described in the section called *Setting Parameters for CYCAD / PC1200 Option*, page 17.
- Import the EX_ORI.DXF file by clicking on the **CAD ➔** function key, or by pressing the **CAD ➔** function key. (See also: *Limitations*, page 15).

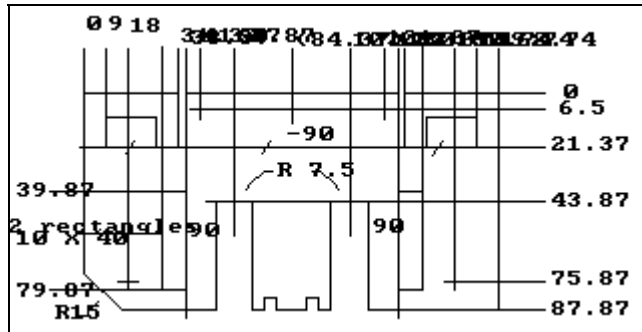


Confirm and the drawing shown hereunder appears.

- Confirm and the drawing shown hereunder appears.

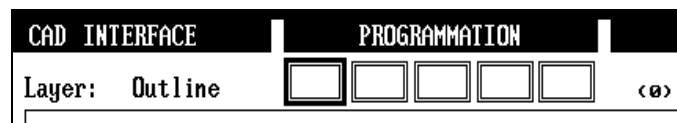


Since no other level has been defined, all elements are located in the **All other** levels. This is indicated by the fact that all lines are blue. It can be seen that in this drawing, only the upper right portion can be used by PC1200. This is the flat layout of the product.



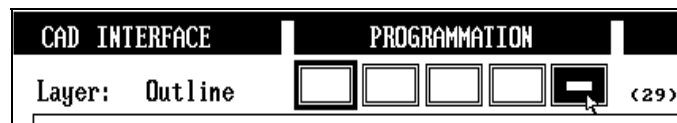
- Zoom into the relevant portion.

- Activate the **Select polyline** icon  and click onto a segment of the outline in order to select it.
- Activate the **Outline** level.

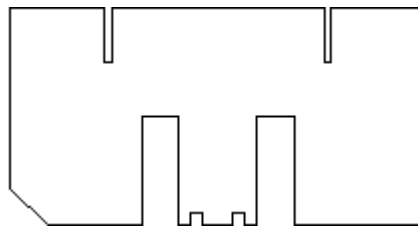


Then click onto the **Transfer** icon , and the outline changes to cyan color.

- To see the contents only of the **Outline** level, mask the **All others** level. It is not necessary to mask other levels because, for the time being, they do not contain any elements.



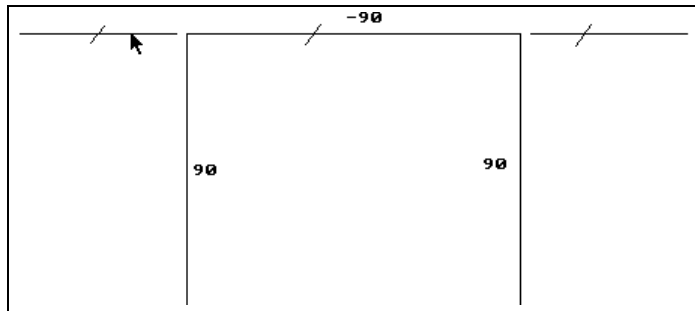
The result in the CYCAD window then looks like this:



At this stage, if no other indication is available, it is possible to transfer this outline into the PC1200 software by clicking the **PRODUCT**  function key. It will, of course, be necessary to add all missing information such as the material, the bends with their values, the tools and their position etc.

- If other information is available, display once again the whole drawing by clicking inside the blue rectangle.
In this example, the bending lines have been specified, as well as the angle values. Activate the **Bends** level, select the bending lines and related texts, and transfer them into the **Bends** level in the same fashion as before.

For this operation, the **Selection**  icon and the **Select Text**  icon, and if necessary the zoom function must be used. The transferred elements will appear in yellow color.



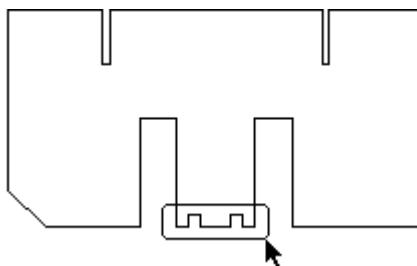
This is what you should now have on your screen after having transferred the bends with their values, and having masked the **Outline** and **All others** levels. The three small oblique lines designate forced bends (see *CAD Drawing Conventions*, page 9).

- Convert the product by clicking on the **PRODUCT**  function key. Caution: the masked levels will not be converted, i.e. if you have masked the **Bends** level, the bends will not be transferred to the current PC1200 product. If the **Outline** level has been masked, an error appears in the moment of conversion.
- In the PC1200 program, add the tools and their positions, and test the feasibility like you would on another product.

SIMPLIFICATION

If the outline of the product contains a large number of small segments which are unimportant for the bending of the product, it may be useful or necessary (max. number of bends reached) to simplify the outline. In the example used, the two small notches of 5 x 5 mm will be used. The distance between the two notches is 12 mm.

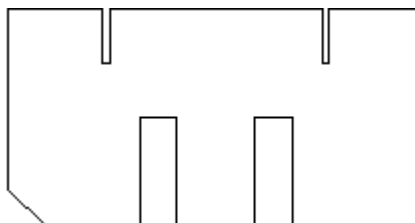
- Call up the **PREFERENCES** window with the  icon.
- Program **Simplification limit size: 13**
(It is suggested that you try out other values in order to understand the way how this functions).
- Activate and display the **Outline** level.



- Activate the **Select window**  icon and select the group of notches.

- Click on the **Simplification**  icon.

Result of the operation:



EXPORTING A FILE IN DXF OR IGES FORMAT

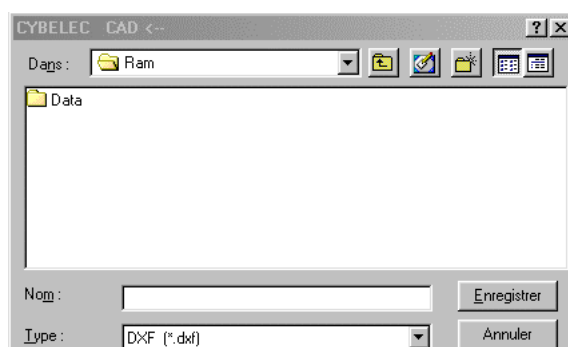
It is possible to export a product created on PC1200 in a DXF or IGES file for use on a CAD system. It is also possible to convert a DXF file into a IGES file, and vice-versa.

The exported file will comply with the CYCAD conventions as described in the *CAD Drawing Conventions* section, page 9.

The file will be exported to the peripheral chosen in the **PREFERENCES** page. In this page, you can also select the export format (DXF or IGES), and the number of segments for each arc.

To export a product from PC1200:

- In the PC1200 program, call up the product to be exported
- In the CAD interface program, call up the current PC1200 product by clicking on the **PRODUCT** ➔ function key.
The product is now displayed in the CYCAD window.
Note: CYCAD only calls up the current product. It is thus not necessary that the product be memorized in PC1200.
- If necessary, you may move or delete elements.
- Click on the **CAD** ⬅ function key.
- Select the directory in which the file will be saved.



- Enter a file name and confirm.

EXAMPLE FILES

On the *CYCAD Example files diskette* several example files are available. Take the necessary time to open them and to have a look at them in your CAD system. This is particularly useful to understand the use of the different levels.

The diskette also contains a number of converted products which can be read directly with PC1200. The numbers **PRODUCT 20000** (to **20055**) designate the corresponding PC1200 product numbers.

The following files can be found on the diskette:

EX_ORI.DWG	Drawing created on AutoCAD in native format.
EX_ORI.DXF	File created on AutoCAD in DXF format from the EX_ORI.DWG file. This is the example file used in the section <i>Importing a CAD File</i> .
EX_CY.DWG	This is a copy of the EX_ORI.DWG file. It has been modified so as to enable automatic conversion. This means that the indications contained in the <i>CAD Drawing Conventions</i> section have been respected.
EX_CY.DXF	File created on AutoCAD in DXF format from the EX_CY.DWG file. See the comments below, and also PRODUCT 20000 .
EXAMPLE1.DXF	DXF file containing the example of a product with two internal cuts. See also PRODUCT 20001 .
EXAMPLE2.DXF	DXF file containing various ways of imposing forced bends. See <i>The "Bends" Level</i> , page 12, and also PRODUCT 20002 .
EXAMPLE3.DXF	DXF file containing an example of internal forced bends (See <i>The "Bends" Level</i> , page 12).
EXAMPLE4.DXF	DXF file containing the example of a product for which the bend order and the supports have been determined on CAD system level. This bend order is recognized by PC1200 (see <i>The "Sections" Level</i> , page 13).
PROB2.DXF to PROB9.DXF	DXF files relating to the <i>Problems and solutions</i> section (page 33).
SUPPLMN2.WRK SUPPLMNT.WRK (DATA directory)	On this diskette are also some sample DXF files already converted.

The EX_CY.DXF file is the same as EX_ORI.DXF, but it has been prepared for CYCAD at CAD system level. When looking at this file with your CAD system, or importing it into CYCAD, you can see that the outline, the bends and the angles have been placed in the levels according to CYCAD "standards", and that information using CYCAD keywords have also been integrated. Because of this, the file can be converted after having been imported directly into a PC1200 product (**PRODUCT** ← key) without having to work on it by means of the CYCAD interface. It will, however, still be necessary to add in PC1200 the missing information and to test the feasibility of the product.

USE OF THE SAMPLE DISQUETTE

In order not to modify your existing environment, the disquette has also been made in such a way that it can be read directly by PC1200. To this effect, the disquette contains the machine parameters and tools making it possible to visualize the converted sample products.

To read information directly on the disquette from PC1200:

- Call the **WELCOME** page.
- Select **ACTIVE PERIPHERALS = FLOPPY**.
(See also: 2D Reference Manual).

In the **CAD INTERFACE** page, access the DXF example files by choosing the **DXF_A** peripheral.

Note: The 3.5" disquette unit must be A:
If this is not the case, you must modify the machine parameters page **PERIPHERALS** accordingly.

To return back to your original configuration, program **ACTIVE PERIPHERALS** in the way they were before.

Normally, the default value is **INTERNAL**.

PROBLEMS AND SOLUTIONS

When using CYCAD, it may happen that the software does not react in the way you are expecting. In such cases, the present section may be useful, for it explains the most frequently problems encountered..

On the CYCAD disquette, in the PROB directory, you will find several DXF example files which may cause problems. They are described in the corresponding sections.

ORDER OF SECTIONS

CYCAD converts the sections in the order of construction. The section established first will be given the number 1, the second the number 2 etc. If there are numbered sections in the CAD drawing, this numbering will be ignored.

The order of the sections influences the bending order, because PC1200 starts with section 1.

PROB2.DXF shows a DXF file where the sections were created in the following order: 1 horizontal, 2 vertical. However, the numbering has been "inverted", i.e. the vertical section was given the number 1, and the horizontal section has number 2.

If you import this example file, you will see after conversion that (**PRODUCT FLAT**) section number 1 is the horizontal and section 2 the vertical section. They do correspond to the order of construction.

The PROB3.DXF file is identical to PROB2, but the vertical section was created first, followed by the horizontal section as a second step.

You can import this example and see the difference in the **PRODUCT FLAT** page.

BENDING LINES

The bending lines must be continuous lines, because on certain CAD systems a "dotted" line is considered to be a multitude of short lines.

It is indispensable that the bending lines touch the outline at both extremities, otherwise the line will not be converted.

The PROB7.DXF file shows a product with a "dotted" bending line. It is not possible to select the line in a single operation, and even if each element is selected, this bending line will not be converted.

The PROB8.DXF file shows a bending line which does not touch the outline (top). During conversion, this line is ignored and will be lost.

Solution:

The only remedy is to make the necessary changes at CAD level.

OPEN OUTLINE

The conventions specify that the outline of a product must be closed. CYCAD does not signal any particular error if this point is not respected. To the eye, an outline may seem to be closed even if it isn't. This problem will show up after conversion in the **PRODUCT FLAT** or **PRODUCT 3D** page.

The PROB4.DXF illustrates this explanation. The outline is not closed in the upper left corner. To see the "gap", it is necessary to zoom extensively, for the opening is only 0.05 mm wide. See also **PRODUCT 20054** on the *CYCAD Example files* disquette.

The PROB6.DXF file provides another example of an unclosed outline. In the upper right corner, the two lines are crossing each other (exaggeratedly for this example) but they have not been declared as closing the outline.

During conversion, the system will close the outline by using the extremity of either one of these lines, and the outline will be incorrect. In this example, a side-effect will be that the bending line no longer touches the outline at one extremity and will therefore not be converted (see Bending Lines).

Solutions:

- 1.- Open the file with a CAD system and correct it by closing the outline.
Note: The outline of the internal cuts must also be closed.
- 2.- Use the functions at disposal:



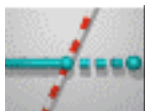
Create segment

Creates a straight segment between 2 lines.
A click close to a point creates the end of the segment on this point. The segment is created in the same level as that of the first line.



Lengthen segment

Click on a line which becomes the support straight line.
Click on another line. This line is lengthened, so that it cuts the support straight line.



Trim two crossing segments

Click on a segment which becomes the support. Click on the segment which crosses. The selected part of the 2nd segment is eliminated up to the intersection with the support segment.

CROSSING BENDS

The conventions do not allow bends to cross each other. Sometimes, the eye cannot distinguish that bends are crossing. The product will be converted without any particular message being issued. The error will be found in the **PRODUCT 3D** page. The product will not be displayed, and the message **Crossing Bends** will be shown. The product can be seen in the **BEND 2D** page.

The PROB5.DXF file illustrates an example with crossing bends. The upper horizontal bend is too low by a value of 0.01 mm. See also **PRODUCT 20055** on the *CYCAD Example files* disquette.

Solution:

It is recommended to correct the problem at its source. However, since the product can be displayed in 2D, this product can be worked on, but only in 2D. Depending on the value of the error, it will be necessary to correct the backgauge axis.

SUPERPOSED LINES

It may happen that you cannot select a line, or that only part of a line is being selected.

Sometimes, during construction of a product, some construction lines remain in the file. They are "hidden" but they may cause you some surprises when selecting or converting items.

The PROB9.DXF file gives a small example of the phenomenon you might encounter.

On the outline, two supplementary lines have been superimposed. One is located on the vertical left line, the other one on the upper horizontal line. The vertical line has the same length as the line it covers, the horizontal one is shorter.

- Call the PROD9.DXF file into the **CAD INTERFACE** page.
- Activate the  icon and try to select the left vertical line. It seems to be impossible to select it.
- Now try to do the same with the upper horizontal line. You will see immediately that it is shorter.

Solution:

Remember that one click activates the selected object, and that the subsequent click de-activates it. The reaction can be different depending on the fact whether the supplementary line is situated on top of or underneath the line of the drawing.

In the case of the vertical line, select the  icon and click **once only** on the vertical line. Even though it is of blue color and does not seem to be

selected, click the eraser  to remove the line.

Proceed in the same way with the horizontal line.

Maybe you are not sure whether you have chosen the right line. In this case, instead of erasing to line, you just transfer it into the **Auxiliary** level. Then you mask it and check the situation.

NOT RECOGNIZED LEVEL

If, after having read a CAD file, the elements are not "placed" automatically in the correct levels, and if the indications **CYCAD : OUTLINE** or **CYCAD : BENDS** (or other similar information) are visible, check the following points:

- **Outline level**
 - Only the outline and possible reserved keywords may be seen in this level.
 - The text CYCAD : OUTLINE must be written on one single line.
 - The text CYCAD : OUTLINE must be written in upper case letters.
 - The double point must be preceded and followed by one space.
 - The syntax of the reserved keywords must be imperatively respected (see the corresponding section).
 - Make sure you use the decimal point and not the comma.
 - The keywords must be placed one on top of the other.
 - Make sure the outline is completely closed.
- **Bends level**
 - This level may only contain bending lines and possible reserved keywords.
 - The text CYCAD : BENDS must be written on one single line.
 - The text CYCAD : BENDS must be written in upper case letters.
 - The double point must be preceded and followed by one space.
 - The syntax of the reserved keywords must be imperatively respected (see the corresponding section).
 - In this level, there must not be any space before or after the equal sign.
 - Make sure you use the decimal point and not the comma.
 - Place the value angles (and the keywords) in a univocal way near the related bending line.
- **Sections level**
 - Same remarks (see corresponding section).

Hint

In any event, if you have doubts about the syntax, you can create a product with the PC1200 software and export in in DXF or IGES format so as to examine it with your CAD system.

NOT CONVERTED ELEMENTS

It may happen that some elements are not converted. In this case, please make sure that:

- All levels are visible (not masked) during conversion.
See *Masking a level*, page 23.
- The element is really located in one of the 3 levels **Outline**, **Bends** or **Sections**.
- Both ends of the bending lines really touch the outline (see *Bending Lines* earlier in this section). If necessary, check both extremities with a strong zoom enlargement.

CAD INTERFACE PAGE NOT AVAILABLE

If the **CAD INTERFACE** page doesn't appear in the main menu:

- Verify that the protection key contains really the option 2. See *Setting Parameters for CYCAD / PC1200 Option*, page 17.
- Verify that the option **CAD INTERFACE Page** in the **OPTIONS** page of the machine parameters has been activated. See *Setting Parameters for CYCAD / PC1200 Option*, page 17.
- This page is available only in the PC1200 software, but not in the numerical control.

This page has been left blank intentionally.

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